Rahul Shah

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Education

University of California, Berkeley — Berkeley, CA

Bachelors in Computer Science; Statistics; Mathematics (Triple Major) GPA: 3.9 — *Jun 2020 – May 2024*

Organizations: CS Honor Society Officer, CSUA (CS Club Executive Officer), CSM (CS 61A/B/70/88 Tutor)

Relevant Coursework: Algos, Compilers, Data Mining*, LinAlg, Multivar Calc., Parallel Systems†, Probability†, Signal Processing, Robotics†, Computer {Architecture, Graphics†, Security, Vision*}, Neural Nets†, Deep RL†, NLP†, Causal Inference*, Forecasting†, Real Analysis*, Financial Engineering*†, Theoretical Stats*† — †= Graduate *= In Progress

Experience

- **Schonfeld (Quantitative Hedge Fund)** — New York, NY
  *Applied Machine Learning Engineer (Alpha **Quantitative Researcher**) Intern* — *Apr 2023 - Aug 2023*
  - Exploratory Data Analysis using **Databricks**' PySpark >100 GB of alternative data from AWS S3 and EC2, with storage optimized via Parquet files, for a statistically backtested signal. Under **Systematic Trading** team (QIS).
  - Wrote an Online Algorithm to improve latency by >50% via Statistical Estimation for a **HFT Stat Arb** strategy. Measured effects of out-of-sample distributional shift via **divergence** measures. Wrote scheduler to ingest live data.
- **Arrowstreet Capital LP** — Boston, MA
  Software Developer *(Quant Dev) Intern @ Arrowstreet's **Research Core** Team* — *Jun 2022 - Aug 2022*
  - Fine-tuned a local **FinBERT** model from **Hugging Face** on over 100,000 vendor filings of various languages & correlated sentiments with word embeddings using **Cosine Similarity**, Jaccard Index, and Minkowski Distances.
  - Wrote distributed code with **Numba** and **Dask** to discover hidden signals for equity alpha research. Made robust improvements to the firm-wide single systematic process with equities toolings for Bayesian Regressions.
- **Electrical Engineering & Computer Sciences @ University of California, Berkeley** — Berkeley, CA
  Teaching: Head TA, uGSI, Tutor, Reader; Researcher, Research Assistant — *Jan 2021 - Present*
  - **Head Teaching Assistant - CS C100: Data Science and Machine Learning**: Managed admin logistics for ∼300 students related to the EdStem communication platform, assignment extensions (automated using Gradescope API), Incomplete Student assignment linkages and DSP communications, held OH and taught sections
  - **Lead Teaching Assistant - EECS 122: Communication Networking/Internet**: Lead all administrative course logistics, **created** all existing course **content**, held office hours and **taught section** on-campus
  - **Reader, Academic Student Employee**: Graded (and wrote) exams on Data Structures (Lists/HashMaps/Heaps/Trees/etc) & Discrete Math (Logic, Crypto, Markov Chains, etc) & held Office Hours for **800+ students**
  - **Berkeley Research - Artificial Intelligence**: Deep Learning & **NLP**: Trained an **LSTM** to predict student grades from attendance and analyzed **Time Series** to determine earnestness with new SOTA **metrics** (Published a paper to be presented at ACM's 2023 conference). Fine-tuned a **LLM** on past TA messages on a Q&A Platform.

Projects

- **GPU-accelerated Factorization Machines**: Boosted recommender systems training speed by **21.4%** using a novel randomized algorithm that capitalized on data **sparsity** to parallelize Stochastic Gradient Descent. Achieved via custom CUDA structures for efficient **low-rank** polynomial kernel approximation. Profiled bottlenecks with VTune's Roofline.
- **Berkeley AI Research (BAIR) Lab**: Collaborated with Simeon Adebola (PhD candidate) and Professor Goldberg from AUTOLab to develop a robust **automated** cable insertion system for **dynamic** environments. Applied CV and flood-fill algorithms for cable & channel identification using perception, segmentation, tracing & endpoint **detection**.

Awards

- **Donald M. Rizzo Memorial Scholarship Recipient**: Awarded from international NPO: Lions Club International
- **Winner of the Congressional App Challenge**: Invited to Capitol Hill to demo mobile app to House of Representative members at #HouseOfCode, the most prestigious student STEM expo, at the U.S. Capitol Building
- **Capture The Flag Security Competitions**: Ranked **top 50** in all of {PicoCTF, HSCTF, BCACTF, CSAW Red}

Skills

- **Languages**: Python, Java, SQL, C, JavaScript, Typescript, Dart, Go, R, C++11, CUDA, OCaml, x86, Lisp, C#
- **Technologies**: PyTorch, Pandas, git, Flutter, React, React Native, Node.js, Firebase, Kubernetes, RegEx, LaTeX
- **Hobbies**: Climbing/Hiking, Reading, LLMs, Trading/Predicting Events, Escape Room, Traveling, Pokemon, Chess